UK Top 50 Playlist Market Structure, Artist Diversity & Content Analysis

Introduction

This student project analyzes daily UK Top 50 playlist snapshots to understand artist concentration, diversity, collaborations, explicit content, release format and track duration.

Dataset and Data Quality

The raw dataset contains 27,800 rows across 555 dates from 18 May 2024 to 27 November 2025. There are 12 exact duplicate rows. One date contains repeated date-position records, so the analysis keeps the first record for each date-position to preserve a daily Top 50 structure.

Methodology

- Dates and text fields were standardized.
- Multi-artist credits were split using the & delimiter from the project brief.
- Collaboration and artist-count features were created.
- Artist concentration was measured using an HHI-style index and a top-five share.
- Explicit content, release type and duration were compared across rank groups and categories.

Key Findings

- 27,750 cleaned playlist entries were used for the analysis.
- 363 artist credits were identified after splitting collaborations.
- The average number of unique artist labels per day was about 39.0.
- Collaborative tracks represented about 18.4% of entries.
- Explicit tracks represented about 32.0% of entries.
- Album entries represented about 60.0%, singles about 39.7%, and compilations about 0.3%.
- Average track duration was about 3.29 minutes.

Recommendations

The dashboard can be used to monitor artist concentration, collaboration activity, content mix and release format across selected dates. These indicators can support UK-focused planning when combined with other audience and commercial information.

Limitations

The provided dataset does not include artist nationality, label, marketing spend, streams or audience demographics. Therefore, domestic-versus-international preference cannot be directly measured from this file alone. Correlation is also not treated as proof of causation.

Conclusion

The project converts daily UK Top 50 snapshots into an interactive market-structure analysis. The Streamlit dashboard makes the main patterns easier to explore without using machine learning, because the assignment is focused on descriptive market analysis.